

Advanced Self-Discharge Measurements of Lithium-Ion Cells and Comparison to Modeling

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Abstract— Lithium-ion batteries (LIBs) are currently the most relevant energy storage solution for a wide field of applications starting from mobile communication and going to high power applications in electric vehicles. To assess the quality of a LIB either during production or in post-production, its self-discharge rate is an important parameter. Here we present a new method for precise potentiostatic self-discharge measurements (SDMs) that is very sensitive and considerably faster than other currently available methods. We validated the new SDM by measuring ten commercial 3000 mAh 21700 LIBs resulting in roughly 3 μ A self-discharge current with a noise level of 0.1 μ A, at 60% state of charge (SoC). Higher self-discharge currents of about 40 μ A were observed at a higher SoC of 80%. To model the SDM, the electrochemical processes are coupled with a 3D temperature FEM and an electric circuit model of the cell self-discharge, resulting in a good overlap and a quantitative value of the LIB self-discharge resistance. Furthermore, we tested a realistic physical self-discharge scenario of the formation of Lithium dendrites by comparing experimental and modelled self-discharge resistance values.

Keywords— *electronic measurements, equivalent electric circuit, finite element method, Lithium ion batteries, modelling, potentiostat, self-discharge.*

I. INTRODUCTION

Lithium-ion batteries (LIBs) are rapidly evolving as the most prominent energy storage solutions due to their high energy densities [1-6]. However, LIBs gradually self-discharge when stored under open circuit conditions [7], with low self-discharge rates when the cells are of high quality and with high self-discharge rates when manufacturing defects occur or when battery degradation is significant. Several studies show that standard LIBs are ranked among the batteries with lowest self-discharge rates [8, 9]. Nevertheless, due to their high importance, rapid growth and strong market impact there is a significant interest in understanding and controlling LIB self-discharge [10]. For example, when cells are assembled into battery packs, differing rates of self-discharge lead to voltage imbalances within a battery pack and short life-time.

The major factors contributing to self-discharge include internal and external current leakage paths, particulate contaminants and dendrite growth producing internal “micro-shorts”, electrolyte degradation, electrode passivation by decomposition and/or film growth on electrodes, and internal pressure build-up. In a recent study the main cause of self-discharge was attributed to parasitic reactions between the

active material and electrolyte [11], resulting in a degradation of the active material and transition metal dissolution. In a study done by Broussely et al. [12] using prototype LIBs, the major contribution to self-discharge was anode corrosion. Kazazi et al. [13] argue that the self-discharge in Li-S batteries is related to the polysulfide shuttle and to the corrosion of the current collectors.

Currently available methods for measuring self-discharge of batteries are relatively time consuming. Such methods include the direct measurement of loss of charge, which involves charging a cell at a desired state-of-charge (SOC), and keeping the cell to rest for a few weeks, then discharging the cell to determine how much charge was lost. An alternative method is based on the measurement of the open circuit voltage (OCV) decay rate [7, 14, 15]; assuming OCV is directly related to the cell’s SOC. This technique involves monitoring OCV over a long period of time, typically two weeks or more. Although these methods can be used to measure self-discharge rates, they are slow and indirect.

Here we describe a new direct potentiostatic method to measure the self-discharge current of batteries, yielding accurate and precise results within a few hours. The new self-discharge measurement (SDM) method is presented and ten commercial LIB cells are tested. The measurements are compared to an extended LIB model including the electrochemistry, the temperature 3D distribution, and an electric circuit model describing the self-discharge resistance and current. Furthermore, a realistic scenario of self-discharge in a LIB due to a conducting Lithium filament is modelled, and the results are discussed.

This paper is organized as follows. Section II introduces the methods for SDM, and for the LIB modeling. Section III discusses the obtained results. The paper is concluded in Section IV.

II. METHODOLOGY

A. Self-discharge measurement (SDM)

An equivalent electric circuit model of self-discharge of LIBs is shown in Fig. 1, right part. In this electric circuit, the voltage of a cell with an effective capacitance C_{eff} drops due to the self-discharge current I_{SD} flowing through the parallel self-discharge resistor R_{SD} , which is typically in the range of $k\Omega$ to $M\Omega$. In the potentiostatic SDM described here (Fig. 1, left part) the initial step is to precisely measure the cell voltage V_{cell} , and to adjust the internal voltage source of the SDM setup for a close match with the LIB within a few

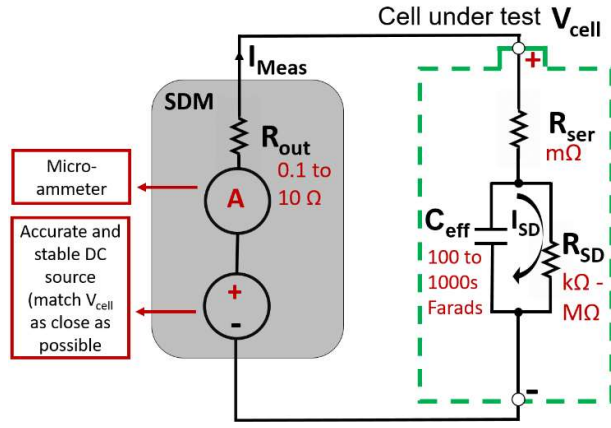


Fig. 1. Self-discharge measurement (SDM) principles. An accurate and stable DC source with an output resistor R_{out} is combined with a sensitive micro-ammeter measuring the current I_{Meas} . The LIB is represented by the cell internal series resistance R_{ser} , the effective cell capacitance C_{eff} , and a parallel resistor R_{SD} providing the self-discharge. The cell open circuit voltage (OCV) is denoted by V_{cell} and the internal self-discharge current by I_{SD} .

micro-volts. The internal source of the SDM equipment (Keysight BT2152A) is constructed for the lowest possible drift and can be considered as ultra-stable ($< 10 \mu V$ drift over 24 hours).

In order to start the actual SDM, the voltage source gets connected to the LIB through an adjustable output resistance R_{out} , usually in the range from 0.1 to 1 Ω . The current I_{Meas} is measured by a micro-ammeter with an accuracy $< 1 \mu A$. During the LIB self-discharge, the voltage source of the SDM starts to supply a current which is exponentially converging

towards an end value, which, once equilibrium is achieved, is equivalent to the measured self-discharge current. The corresponding time constant of the cell under the SDM measurement is $\tau = (R_{out} + R_{ser}) \times C_{eff}$, depending on the cell's effective capacity C_{eff} and on the total series resistances. In SDM, a lower output resistor R_{out} leads to faster convergence and faster measurement of the self-discharge current. However, it also increases undesired artefacts and noise contributions because cell voltage fluctuations (e.g. due to temperature changes) are directly converted into current fluctuations with a gain proportional to $1/R_{out}$. The best trade-off between measurement speed of the self-discharge on the one hand, and the corresponding noise on the other hand depends on many factors and is not further examined in this work. Here, the following parameters are considered for the SDM: an initial current I_{Meas} of 0 A, an output resistance R_{out} of 1 Ω , and a measurement sampling interval set to 10 s. SDMs were performed in a temperature chamber set at 23 $^{\circ}C$. Fluctuations of the chamber temperature due to feedback from the heating control are effectively low-pass filtered by wrapping the electrical fixture with thermal insulation material, and by including thermal mass elements next to it. The temperature was measured with miniature PT100 sensors mounted directly on the cells using thermal grease. The readout was performed over a 7.5 digit DMM (Keysight 34470A) in the resistance range. Before starting the SDMs, the cells were left to rest in the temperature chamber for 24 hours at about 60% SOC with a cell voltage $V_{cell} = 3.55 V$. This procedure kept the overall temperature changes directly at the cell below 0.04 $^{\circ}C$ during the 25 hours SDM. These measurement conditions were optimised in order to facilitate comparison to the simulation results.

The SDM concept maintains a constant voltage on the cell under test (i.e. the initial OCV sensed by the SDM at the start

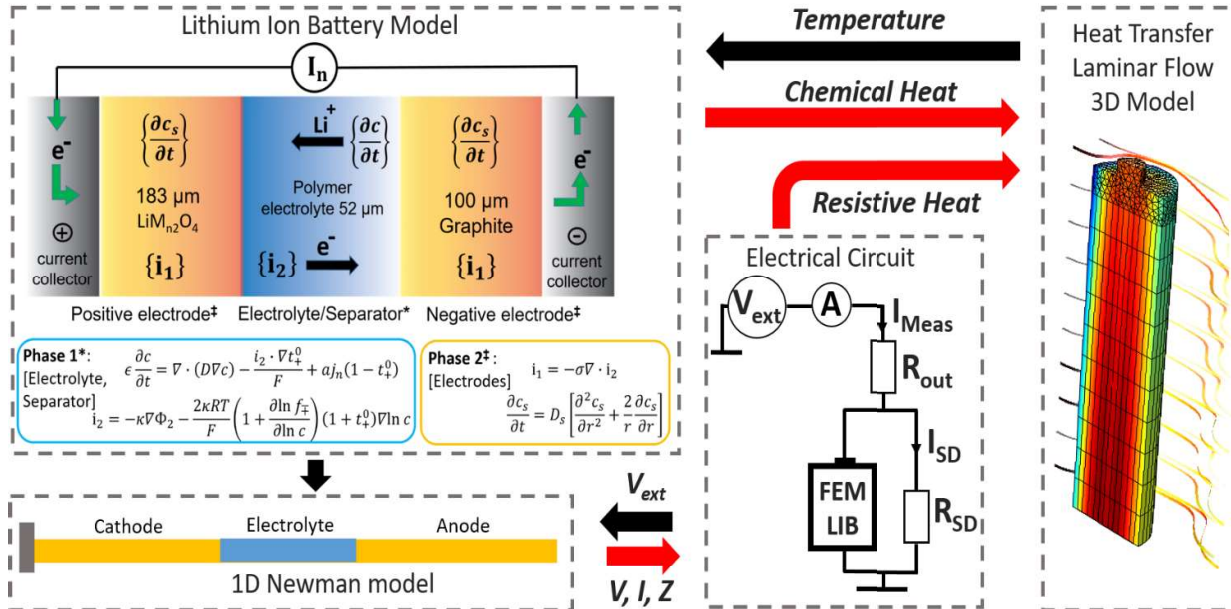


Fig. 2. Lithium-ion battery (LIB) modelling. Left: The 1D Newman model is used to describe the electrode interface processes including the electrochemistry of the anode, the cathode, and the electrolyte/separator. Center: Electrical circuit model of the self-discharge process. Right: The heat that is generated in the 1D Newman model is coupled with a 3D FEM temperature model. The 3D FEM model of the cell is shown including meshing and the laminar flow cooling. The relations between the 1D Newman model, the electric circuitry model of the self-discharge, and the 3D temperature model are illustrated by arrows including the physical parameters. Heat, voltage, and current are outputs of the 1D Newman model and the temperature and voltage are inputs.

of the measurement) and no load is connected to the cell, thus nonlinear behaviour due to voltage change of the cell does not occur. In addition, different cell effective capacitance values change the total time constant values but do not influence the end values of the self-discharge currents measured after equilibrium is reached.

A second experiment was done to investigate the effect of the cell state of charge (SoC) on the self-discharge current. The SoC of the 10 cells were incremented up to 80% SoC (with about 3.94 OCV). This was done by first charging the cells fully to 100% SoC using CCCV (constant current, constant voltage) method and then discharging the cells down to 80% SoC. The same ambient conditions were maintained and the experimental results of the self-discharge currents are shown in Fig. 3d. The self-discharge measurement results of both experiments show that with a higher SoC a higher self-discharge current is observed, in the context of a constant temperature. In addition, charge redistribution and longer rest periods of the cells have a strong effect on the self-discharge currents, as can be seen from the first experiment (Fig. 3d), where the cells were stored in room temperature for a long period of time before conducting the SDM experiment. One likely mechanism for the increase of the self-discharge current at higher SoC is that lithium intercalation into the electrode increases the internal pressure (see section IV for further discussion).

B. LIB modeling

The LIB was modelled as an electrical circuit coupled with a 1D electrochemical Newman model and a 3D finite element model (FEM) of the cell temperature (Fig. 2). FEM was carried out with COMSOL Multiphysics 5.4, using the electrochemical toolbox. A 1D-Newman model [16] was used to describe the electrochemical processes, with the main equations of this 1D-Newman model given in Fig. 2, left panel. They describe the non-isothermal charge/discharge process including the Lithium material balance in the electrolyte. Here, c and ϵ are concentration and volume fraction of the polymer in the cathode, i_2 is the current flowing in the electrolyte, i_1 is the current density in the electrodes, and I_n is the superficial current density of the cell. This 1D electrochemistry model considers a LiMn_2O_4 cathode with 183 μm length, a graphite anode with 100 μm length, and a polymer electrolyte of 52 μm length. As indicated by the arrow, the calculated heat generation in the 1D-Newman model is coupled with a 3D temperature FEM of the cell (Fig. 2, right panel). The self-discharge current is modelled based on the electric circuit shown in Fig. 2, center panel, consisting of an external voltage source V_{ext} , an ammeter, a series output resistor R_{out} , and a parallel self-discharge resistor R_{SD} . The external voltage source V_{ext} keeps the battery at its initial potential value. The main output of this electric circuitry model is the current I_{Meas} flowing through the output resistor R_{out} , which, is then compared to the measured self-discharge current.

As shown in Fig. 5, the 1D model was extended to a 2D axisymmetric model to include a metallic Li dendrite filament as one of the possible reasons of self-discharge. The Lithium filament was positioned in the centre of the cell, with a conductivity of 10 MS/m, bridging the positive and the negative electrode sides. The half-circle interface region between dendrite and porous electrode was modelled as being

electrochemically inactive. Similar as before, the LIB physics was coupled with the heat flow and the electrical circuit to calculate I_{Meas} for various filament radii.

III. RESULTS

As described in the previous section, the SDM measures the current that needs to be supplied to a cell in order to keep its voltage constant. In equilibrium, this SDM current is equivalent to the LIB's internal self-discharge current. Equilibrium is, depending on the output resistance chosen, approached exponentially with a time constant ranging from minutes to hours. Fig. 3 displays the measured self-discharge currents of ten commercial 3000 mAh 21700 LIBs all logged at the same time during a 25-hour measurement. The battery cells were placed in a 6x6 fixture (Fig. 3b), connected to the SDM setup through a printed circuit board (Fig. 3a). The fixture with the cells was installed inside a temperature chamber to keep the temperature constant (see inset in Fig. 3c). The self-discharge currents I_{Meas} show the typical exponentially converging behaviour towards a self-discharge I_{SD} end value (Fig. 3c). For high-quality performing cells the self-discharge current is very small around 1-10 μA (Fig. 3c),

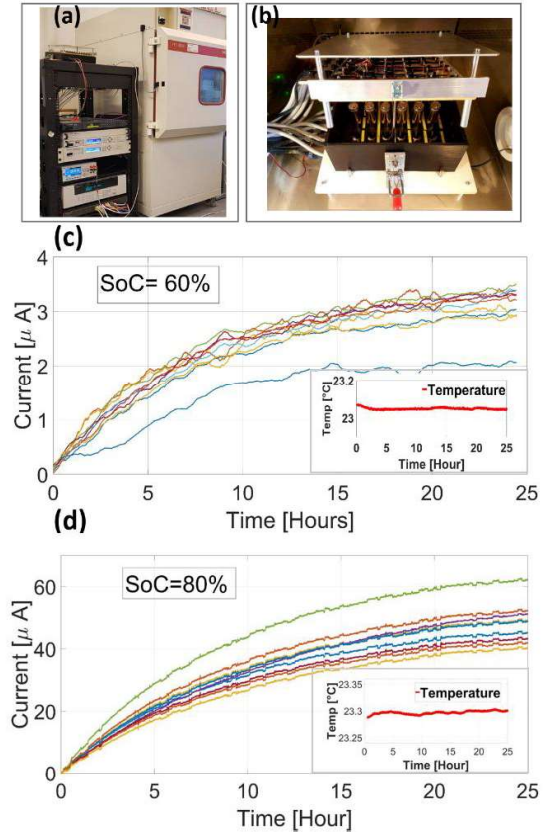


Fig. 3. SDM results. a) The SDM setup including the fixture and the temperature-controlled chamber. b) A fixture is shown for electrically contacting 6x6 LIB 21700 cells, mounted in a temperature chamber. c) Self-discharge current versus time for a 25-hour experiment and ten 21700 cells at 60% SoC. The inset shows the temperature logged during the measurement resulting in very low temperature variations. D) Self-discharge current versus time for a 25-hour experiment ten 21700 cells at 80% SoC.

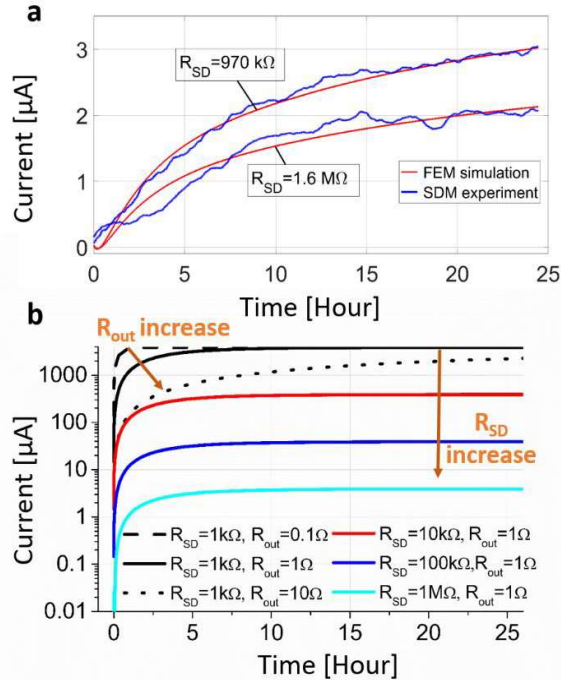


Fig. 4. Comparison of SDM and modelling. a) The measured self-discharge curves are shown for two cells (at 60% SoC) overlaid with the simulation results for the 25-hour experiment. The two SDM curves are fitted with FEM using two different values of the parallel self-discharge resistor (R_{SD}). b) Systematic variation of simulation parameters with the self-discharge current shown for different parameters of the self-discharge resistor (R_{SD}) and the output resistor (R_{out}).

while for bad cells that have imperfections and large leakage the self-discharge current can raise to $100+ \mu\text{A}$ (not shown). For the ten selected LIBs, most of the cells show a similar and very low self-discharge current of $3.2 \mu\text{A}$, while only one cell has even a lower (i.e. better) self-discharge current of $2.0 \mu\text{A}$, with a noise floor of the SDM below $0.1 \mu\text{A}$. In the LIB electrical circuitry model the self-discharge current is represented by a parallel self-discharge resistance that ranges from $\text{k}\Omega$ to $\text{M}\Omega$. Thereby, high performing cells with low self-discharge current have large $\text{M}\Omega$ self-discharge resistance, and vice versa. Both quantities, the self-discharge current I_{SD} and the self-discharge resistance R_{SD} represent important quality criteria of cells that allow to classify them and to sort out suspicious ones during production.

For quantitative interpretation of SDM we modelled the entire measurement setup numerically. Here, the electric circuitry of the LIB, including the parallel self-discharge resistance, is interfaced with a 1D electrochemical Newman model, and the heat source generated from the 1D-Newman model is coupled with a 3D temperature FEM of the cell. Fig. 4a shows the comparison of the model with SDMs for two select cells. A good match is obtained for both cells along the 25-hour time frame using self-discharge resistor R_{SD} values of $970 \text{ k}\Omega$ and $1.6 \text{ M}\Omega$, respectively, which are typical values for high-performing intact cells. In Fig. 4b the model is solved for a set of parameters $R_{SD} = \{1 \text{ k}\Omega, 10 \text{ k}\Omega, 100 \text{ k}\Omega, 1 \text{ M}\Omega, 10 \text{ M}\Omega\}$, and $R_{out} = \{0.1 \Omega, 1 \Omega, 10 \Omega\}$. In this systematic evaluation, by keeping R_{out} constant at 1Ω and increasing the value of R_{SD} , the self-discharge current I_{SD} drops monotonously, representing a low self-discharge cell.

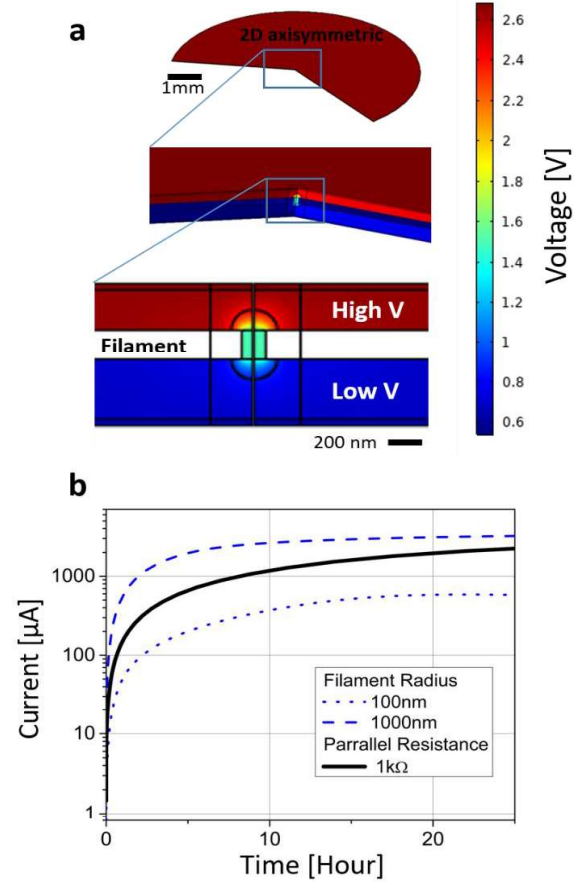


Fig. 5. FEM self-discharge simulation of a microscopic filament. a) The potential distribution of a cell in a 2D axisymmetric FEM is shown with a z-scale ranging from 0.6 to 2.6 V and a lateral scale-bar of 1mm. The filament is shown as a physical source of the self-discharge. b) The modelled self-discharge current plotted versus time. The blue lines represent two different filament sizes, while the solid black curve represents the FEM simulation results using a self-discharge resistor R_{SD} of $1 \text{ k}\Omega$.

Similarly, by keeping R_{SD} constant at $1 \text{ k}\Omega$ and increasing R_{out} a higher time constant of the exponential shape is evident, which is due to the increase of the total series resistance, $\tau = (R_{out} + R_{ser}) \times C_{eff}$.

In Fig. 5 we extended the 1D model to a 2D axisymmetric model, which allows to evaluate how specific LIB defects in materials or physical-chemical processes lead to an increased self-discharge current. We modelled a specific cause of self-discharge which is the formation of Lithium dendrites within the separator of a LIB (Fig. 5.a), and we investigated which impact it has on the self-discharge parallel resistance (Fig. 5b). The simulated SDM curves are based on two different values of the thickness of the filament, 100 nm and 1000 nm. Both curves show a similar shape compared to a parallel resistor of $1 \text{ k}\Omega$. On the one hand this result shows that the effective model of the self-discharge as a parallel resistor is a good approximation, while on the other hand we note that the obtained self-discharge currents, even for the smaller 100 nm filament, are too high in comparison to the experimental curves of an intact LIB. Further modelling of smaller filaments and additional defects will be done in order to better understand the microscopic processes that lead to self-discharge of LIBs.

IV. CONCLUSION

This paper presents a potentiostatic method for self-discharge measurement in batteries. The cell self-discharge is directly and accurately measured within a small fraction of the time required by traditional approaches. The method was validated by conducting two 25-hour experiments with ten commercial 3000 mAh NMC cells; one experiment at a 60% SoC with a long rest period and the second experiment at 80% SoC. The self-discharge current dependency on SoC is related to several physical-chemical causes. Ricketts et al. show that the diffusion increases strongly with increased voltage [17]. Andreas et al. argue that when holding the electrodes at a higher energy state will naturally lead to a relaxation to a lower energy state by any means possible [18]. Iglesias et al. in a study of a capacity fade model showed experiments of measuring self-discharge current through coulombs counting method where the capacity loss of the batteries were much higher at higher SoC, they point out that with higher stress factors of cells (e.g. at higher SoC) capacity fade is more evident [19]. These conclusions concede with the self-discharge current results observed in both our experiments.

In addition, FEM modelling of the SDM was performed. A 1D-Newman model was used to describe the electrochemical processes of LIB, is coupled with a 3D temperature FEM of the cell and an electric circuit model of cell self-discharge. Results from the SDM method and the FEM simulation were compared and discussed. Furthermore, the 1D-Newman FEM model was extended into a 2D axisymmetric one, allowing us to model a realistic self-discharge scenario of the formation of Lithium dendrites within the separator of a LIB. Our findings demonstrate the effectiveness of the proposed SDM FEM model. However, they also demonstrate that even for filament sizes of 100nm, simulated currents are still too high to be the reason for the here measured SDM currents of the tested cells. Future work will therefore involve FEM modelling investigation of alternative physical-chemical and/or production related mechanisms of self-discharge in battery cells.

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